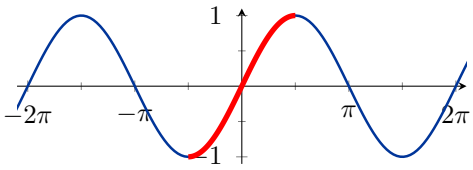
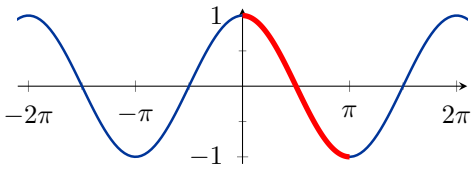
6.3: Inverse Trigonometric Functions

Note that $\sin(x)$ and $\cos(x)$,

- take in the measure of an angle, and
- return the ratio of the sides.

Recall that trigonometric functions are periodic:





Consider the following restricted domains and associated ranges:

	Domain	Range
$y = \sin(x)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$[-1, 1]$
$y = \cos(x)$	$[0, \pi]$	$[-1, 1]$
$y = \tan(x)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$(-\infty, \infty)$

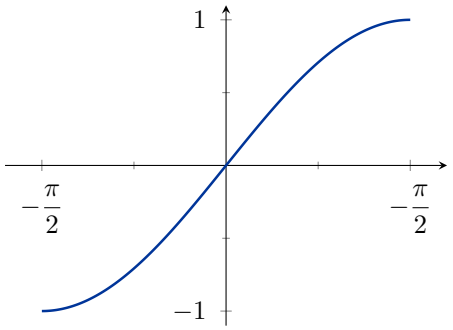
When we use the restricted domains, then these functions are one-to-one. The resulting inverse functions are:

$$\arcsin(y) = \sin^{-1}(y)$$
$$\arccos(y) = \cos^{-1}(y)$$
$$\arctan(y) = \tan^{-1}(y)$$

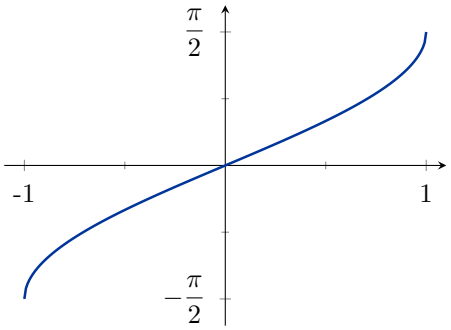
Example. Since $\sin\left(\frac{5\pi}{12}\right) \approx 0.96593$, what is $\sin^{-1}(0.96593)$?

$\sin^{-1}(0.96593) = \frac{5\pi}{12}$

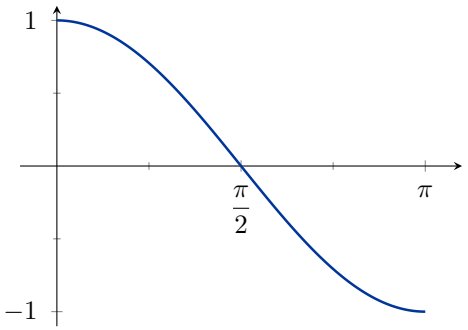
$y = \sin(x)$



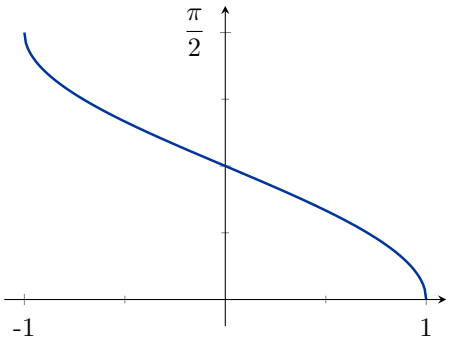
$y = \sin^{-1}(x)$



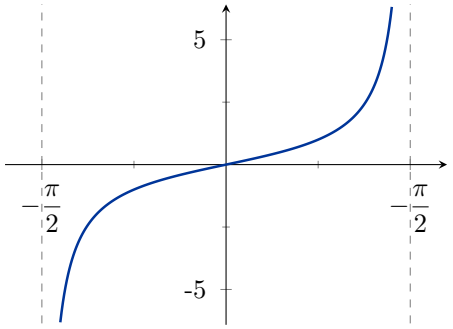
$y = \cos(x)$



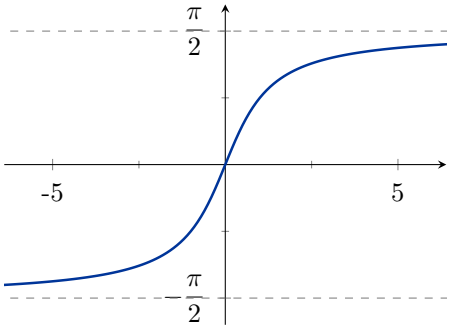
$y = \cos^{-1}(x)$



$y = \tan(x)$



$y = \tan^{-1}(x)$



Example. Evaluate the following:

$\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4}$

$\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$

$\tan^{-1}(-1) = -\frac{\pi}{4}$

$\tan^{-1}\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6}$

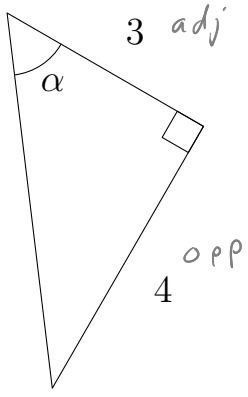
$\frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}} = \frac{1/2}{\sqrt{3}/2}$

Example. Use inverse trigonometric functions to solve for the following angles:

TOA

$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$

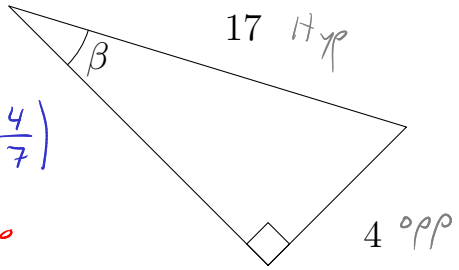
$\approx 53.13°$



SOH

$\beta = \sin^{-1}\left(\frac{4}{17}\right)$

$\approx 13.61°$

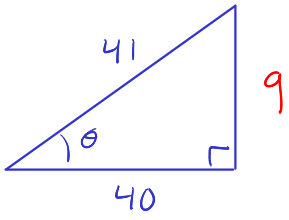


Example. Evaluate the following:

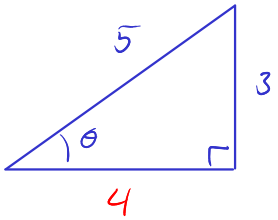
$\sin^{-1}\left(\sin\left(37°\right)\right) = 37^{\circ}$

$\cos^{-1}\left(\cos\left(\frac{9\pi}{4}\right)\right) = \frac{\pi}{4}$

$\sin\left(\cos^{-1}\left(\frac{40}{41}\right)\right) = \frac{9}{40}$

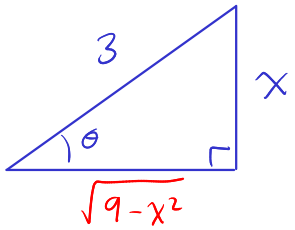


$\tan\left(\sin^{-1}\left(\frac{3}{5}\right)\right) = \frac{4}{5}$



Example. Simplify the following:

$\cos\left(\sin^{-1}\left(\frac{x}{3}\right)\right)$, where $-3 \le x \le 3$



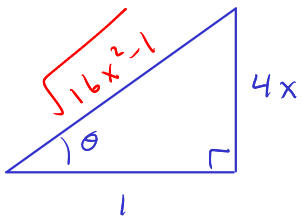
$x^2 + b^2 = 3^2$

$b^2 = 9 - x^2$

$b = \sqrt{9-x^2}$

$\cos\left(\sin^{-1}\left(\frac{x}{3}\right)\right) = \frac{\sqrt{9-x^2}}{3}$

$\sin\left(\tan^{-1}\left(4x\right)\right)$



$(4x)^2 + 1^2 = c^2$

$\sqrt{16x^2-1} = c$

$\sin\left(\tan^{-1}\left(4x\right)\right) = \frac{4x}{\sqrt{16x^2-1}}$